📄 RAPPORT — MINI PROJECT 4

Sécurité des Réseaux et Tests d'Intrusion

1. Introduction

Ce projet a pour objectif de mettre en pratique les concepts fondamentaux de la sécurité des réseaux et des tests d'intrusion. Il couvre la configuration d'un pare-feu, la mise en place d'un IDS, l'utilisation de Nmap pour la reconnaissance, ainsi que la simulation d'une attaque simple.

---

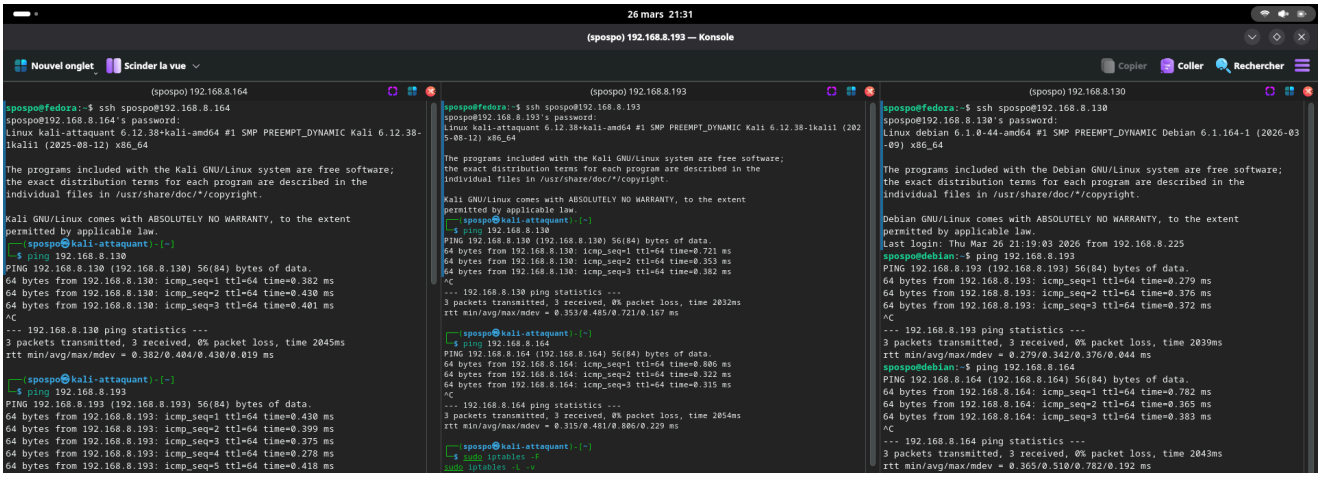
2. Environnement

Trois machines virtuelles ont été utilisées :

Machine	Rôle	IP
Kali Linux 1	Attaquant	192.168.8.164
Kali Linux 2	Analyseur	192.168.8.193
Debian	Victime	192.168.8.130

Les machines sont connectées sur un même réseau interne.

Screens à insérer :



---

3. Partie 4A : Configuration du Firewall

Étapes réalisées :

1. Suppression des règles existantes :

```
sudo iptables -F
```

2. Définition de la politique par défaut :

```
sudo iptables -P INPUT DROP
sudo iptables -P OUTPUT ACCEPT
```

3. Autorisation du SSH :

```
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

4. Autorisation du loopback :

```
sudo iptables -A INPUT -i lo -j ACCEPT
sudo iptables -A OUTPUT -o lo -j ACCEPT
```

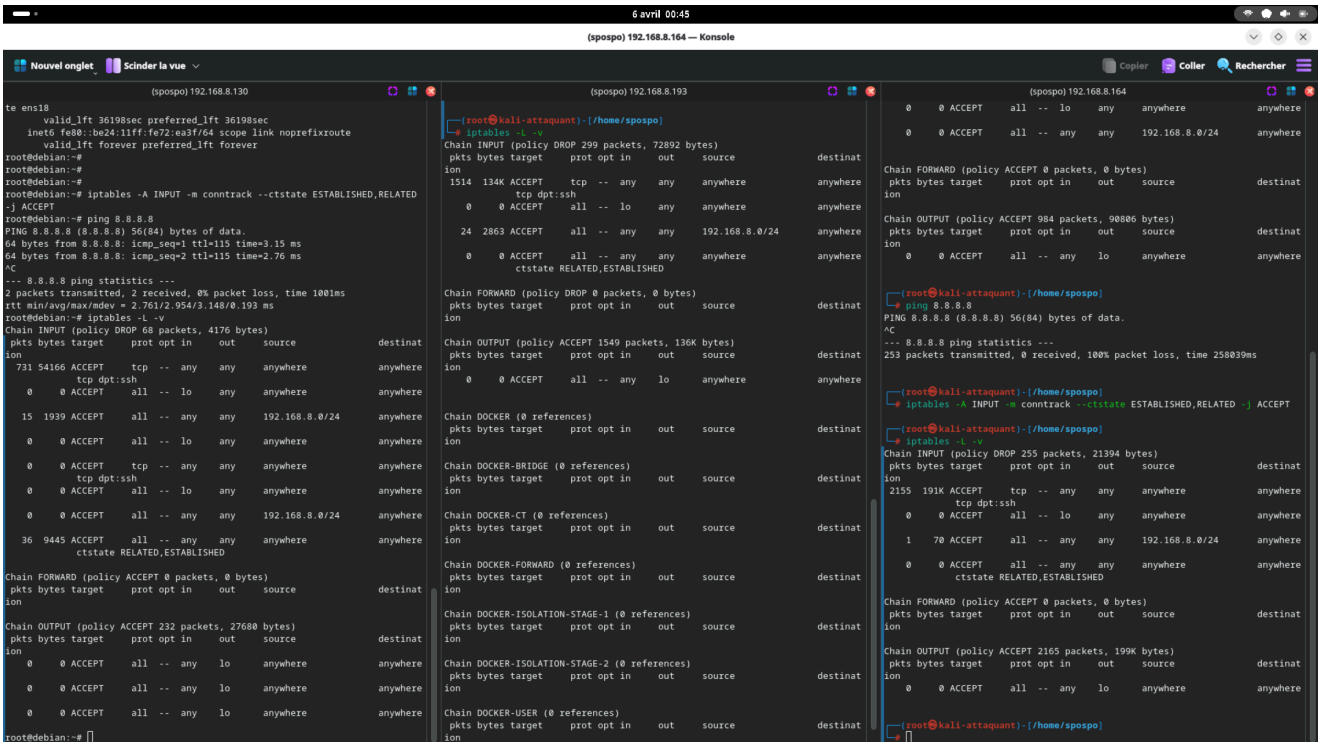
5. Autorisation du réseau local :

```
sudo iptables -A INPUT -s 192.168.X.0/24 -j ACCEPT
```

Analyse :

Une politique restrictive a été appliquée afin de limiter les connexions entrantes. Seuls les services nécessaires ont été autorisés.

Screens à insérer :



---

4. Partie 4B : Mise en place d’un IDS (Snort)

Installation :

```
sudo apt update && sudo apt install snort
```

Configuration :

Modification du fichier /etc/snort/snort.conf :

- HOME_NET = réseau local
- EXTERNAL_NET = !$HOME_NET

Lancement :

```
sudo snort -dev -i eth0 -c /etc/snort/snort.conf
```

Génération d’attaque :

```
nmap -p 1-1000 <IP Ubuntu>
```

Consultation des logs :

```
sudo tail -f /var/log/auth.log
```

Analyse :

Snort a détecté une activité suspecte correspondant à un scan de ports.

Screens à insérer :

- Configuration snort.conf

```
(spospo) 192.168.8.193
  GNU nano 8.6                    /etc/snort/snort.lua
HOME_NET = '192.168.8.0/24'

EXTERNAL_NET = '!$HOME_NET'

include = 'local.rules'

ips =

{

   variables = default_variables,


   rules = [[

       include /etc/snort/local.rules

   ]]

}


alert_json =

{

   file = true

}

                          [ Lecture de 29 lignes ]
^G Aide        ^O Écrire      ^F Chercher    ^K Couper      ^T Exécuter
^X Quitter     ^R Lire fich.  ^\ Remplacer   ^U Coller      ^J Justifier
```

(spospo) 192.168.8.193

```
  GNU nano 8.6                    /etc/snort/local.rules
alert icmp any any -> any any (msg:"PING detecte"; sid:1000001; rev:1;)
alert tcp any any -> 192.168.8.130 1:1000 (msg:"NMAP Scan detecté"; flags:S>
```

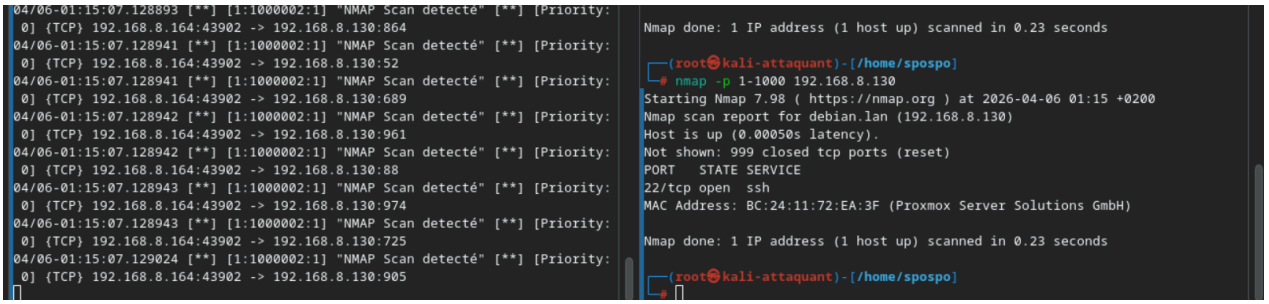
- Snort en cours d'exécution

```
(spospo) 192.168.8.193
└─# nano /etc/snort/snort.lua

┌──(root㉿kali-attaquant)-[/home/spospo]
└─# nano /etc/snort/local.rules

┌──(root㉿kali-attaquant)-[/home/spospo]
└─# sudo snort -c /etc/snort/snort.lua -i eth0 -A alert_fast
--------------------------------------------------
o")~   Snort++ 3.12.1.0
--------------------------------------------------
Loading /etc/snort/snort.lua:
        ips
        host_tracker
        packets
        search_engine
        process
        active
        alerts
        daq
        decode
        host_cache
        alert_json
        output
        network
        hosts
Finished /etc/snort/snort.lua:
Loading ips.rules:
Loading /etc/snort/local.rules:
Finished /etc/snort/local.rules:
Finished ips.rules:
--------------------------------------------------
ips policies rule stats
              id  loaded  shared enabled    file
               1       2       0       2    /etc/snort/snort.lua
--------------------------------------------------
rule counts
       total rules loaded: 2
               text rules: 2
            option chains: 2
            chain headers: 2
--------------------------------------------------
port rule counts
             tcp     udp    icmp      ip
     any       0       0       1       0
     dst       1       0       0       0
   total       1       0       1       0
--------------------------------------------------
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth0
```

- Résultat Nmap & logs avec alertes



---

5. Partie 4C : Scan avec Nmap

Commandes utilisées :

Scan SYN :

```
nmap -sS <IP>
```

Scan ports spécifiques :

```
nmap -p 22,80,443 <IP>
```

Scan UDP :

```
nmap -sU <IP>
```

Détection de services :

```
nmap -sV <IP>
```

Détection OS :

```
nmap -O <IP>
```

Scan complet :

```
nmap -sS -sV -O -p 1-65535 <IP>
```

Analyse :

Les scans ont permis d'identifier les ports ouverts ainsi que les services actifs. Ces informations sont essentielles pour identifier des vulnérabilités potentielles.

Screens :

```
Nouvel onglet  Scinder la vue                                                    Copier  Coller  Rechercher
(spospo) 192.168.8.164
┌──(root㉿kali-attaquant)-[/home/spospo]
└─# nmap -sS 192.168.8.130
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 01:15 +0200
Nmap scan report for debian.lan (192.168.8.130)
Host is up (0.00016s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: BC:24:11:72:EA:3F (Proxmox Server Solutions GmbH)

Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds

┌──(root㉿kali-attaquant)-[/home/spospo]
└─# nmap -p 22,80,443 192.168.8.130
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 01:16 +0200
Nmap scan report for debian.lan (192.168.8.130)
Host is up (0.00031s latency).

PORT    STATE  SERVICE
22/tcp  open   ssh
80/tcp  closed http
443/tcp closed https
MAC Address: BC:24:11:72:EA:3F (Proxmox Server Solutions GmbH)

Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds

┌──(root㉿kali-attaquant)-[/home/spospo]
└─# nmap -sU 192.168.8.130
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 01:16 +0200


┌──(root㉿kali-attaquant)-[/home/spospo]
└─# nmap -sU 192.168.8.130
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 01:24 +0200


┌──(root㉿kali-attaquant)-[/home/spospo]
└─# nmap -sV 192.168.8.130
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 01:26 +0200
Nmap scan report for debian.lan (192.168.8.130)
Host is up (0.000078s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 9.2p1 Debian 2+deb12u7 (protocol 2.0)
MAC Address: BC:24:11:72:EA:3F (Proxmox Server Solutions GmbH)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.51 seconds

┌──(root㉿kali-attaquant)-[/home/spospo]
└─# nmap -O 192.168.8.130
```

```
Nouvel onglet  Scinder la vue                                                    Copier  Coller  Rechercher
(spospo) 192.168.8.164
└─# nmap -O 192.168.8.130
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 01:26 +0200
Nmap scan report for debian.lan (192.168.8.130)
Host is up (0.00035s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: BC:24:11:72:EA:3F (Proxmox Server Solutions GmbH)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.56 seconds

┌──(root㉿kali-attaquant)-[/home/spospo]
└─# nmap sS -sV -O -p 1-65535 192.168.8.130
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 01:26 +0200
Failed to resolve "sS".
Nmap scan report for debian.lan (192.168.8.130)
Host is up (0.00038s latency).
Not shown: 65534 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 9.2p1 Debian 2+deb12u7 (protocol 2.0)
MAC Address: BC:24:11:72:EA:3F (Proxmox Server Solutions GmbH)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.98 seconds

┌──(root㉿kali-attaquant)-[/home/spospo]
└─# nmap -sS -sV -O -p 1-65535 192.168.8.130
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 01:27 +0200
Nmap scan report for debian.lan (192.168.8.130)
Host is up (0.00037s latency).
Not shown: 65534 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 9.2p1 Debian 2+deb12u7 (protocol 2.0)
MAC Address: BC:24:11:72:EA:3F (Proxmox Server Solutions GmbH)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

---

6. Partie 4D : Attaque simple

Reconnaissance :

```
nmap -sS -sV -p 1-65535 <IP>
```

Attaque SSH (Hydra) :

```
hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://<IP>
```

Analyse :

Une attaque par brute force a été tentée sur le service SSH. Cette technique exploite des mots de passe faibles.

Screens à insérer :

- Scan initial

```
Nmap done: 1 IP address (1 host up) scanned in 3.98 seconds

┌──(root㉿kali-attaquant)-[/home/spospo]
└─# nmap -sS -sV -O -p 1-65535 192.168.8.130
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 01:27 +0200
Nmap scan report for debian.lan (192.168.8.130)
Host is up (0.00037s latency).
Not shown: 65534 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 9.2p1 Debian 2+deb12u7 (protocol 2.0)
MAC Address: BC:24:11:72:EA:3F (Proxmox Server Solutions GmbH)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikro
tik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS
7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at h
ttps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.86 seconds
```

- Hydra en cours & résultat attaque

```
┌──(root㉿kali-attaquant)-[/usr/share/wordlists]
└─# hydra -l spospo -P /tmp/test.txt -t 4 ssh://192.168.8.130
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use i
n military or secret service organizations, or for illegal purposes (this is
 non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-06 01
:34:29
[DATA] max 3 tasks per 1 server, overall 3 tasks, 3 login tries (l:1/p:3), ~
1 try per task
[DATA] attacking ssh://192.168.8.130:22/
[22][ssh] host: 192.168.8.130   login: spospo   password: 1234
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-06 01
:34:32
```

---

7. Partie 4E : Méthodologie Pentest

Un test d’intrusion suit plusieurs étapes :

1. Reconnaissance
2. Scan
3. Exploitation
4. Maintien de l’accès
5. Reporting

Un pentest est une activité légale réalisée avec autorisation, contrairement au hacking.

---

8. Conclusion

Ce projet a permis de comprendre l’importance de la sécurité réseau et des outils de détection. Il montre également que des vulnérabilités simples peuvent être exploitées si les systèmes ne sont pas correctement sécurisés.

Des améliorations possibles incluent l’utilisation d’un IPS et l’approfondissement des techniques d’exploitation.